

generators, one for the training set and the other one for our validation set. Then follow the steps:

1. First, we need to normalise the data by passing the rescale option to the ImageDataGenerator object. The rescale=1/255 is a very important parameter because it normalises the image pixel values to have zero mean and standard deviation of one. This helps the model to learn and update its parameters efficiently.
2. Next, we will perform the Image augmentation options that randomly apply some transformation to the Images using ImageDataGenerator object. This will augment our data set and thus helps in improving generalisation.



Figure : Example of an augmented image

3. Create an ImageDataGenerator object for our validation set also but do not perform data augmentation here since it is only for validation purposes and not training. Perform only rescale operation.

Since we have the ImageDataGenerator objects, let us now create python generators from them by passing our training and validation data set. Follow these steps:

1. First, call the .flow () method on the data generators that we created above by passing in the data and label set. X_train and y_train are the



Figure : Creating Python generators for the image dataset